

An electro-mechanical contact model for particulate systems

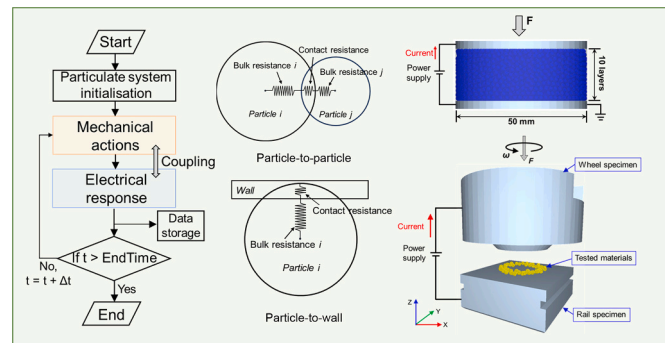
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HIGHLIGHTS

- An electro-mechanical coupled contact model for particulate systems is presented.
- Verification, validation, and application of the contact model are shown.
- Potential and current intensity distributions of particulate system are analysed.
- An index is proposed to evaluate the degree of electrical activity.

GRAPHICAL ABSTRACT



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ABSTRACT

This study presents an electro-mechanical contact model to investigate the electrical response within a particulate system under mechanical loading. The model considers the electrical resistance across particle-to-particle and particle-to-wall contacts by formulating bulk and contact resistances, separately. Hertz and Holm's models are utilised for the contact resistance in the overlapping region. The bulk resistance for a particle is estimated considering particle geometry transformations during the compression. The proposed electro-mechanical contact model is verified and validated against published analytical solutions and experimental data. Finally, the contact model is applied to simulate a high-pressure torsion test to investigate the influence of three different materials on the electrical conduction properties of the system after particle fragmentation. The intensity distributions of electric potential, current and resistance of the granular system are analysed at both particle and system scales. An electroactivity index is proposed for evaluating the contribution of each fragment sieve cut on the bulk electrical conduction behaviour.

1. Introduction

Electro-mechanical coupling in granular materials is both widely reported [1–3] and widely used across a range of industries. For example, optimising compaction parameters of metal powders through

electrical responses [4]; identifying mechanical component defects based on abnormal electrical resistance values [5], and determining train locations through electrical signals from track circuits [6], all involve the coupling between mechanical behaviour and electrical response. Beyond these industrial applications, electro-mechanical

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coupling has recently been applied in several other fields, including the calendaring of battery materials [7], human skin tissue wound healing [8], eco-friendly device development [9], and the production of new polymers [10,11]. To date, extensive experiments have enhanced our understanding of the interaction between mechanical behaviour and electrical response [12–17]. However, experimental studies suffer from high time and economic costs, as well as the difficulty of acquiring data on both particle behaviour and electro-mechanical coupling mechanisms at the grain-scale.

Numerical simulation techniques, known for their efficiency and cost-effectiveness, are increasingly adopted for studying electro-mechanical coupling in particulate systems [18–21]. Among these techniques, the discrete element method (DEM) [22], recognised for its robust capability to analyse particle dynamics, is widely employed by researchers for simulating the electrical transfer characteristics of particles under mechanical actions [23–25]. Renouf and Fillot [26] employed Hertz's theory [27] to determine the overlapping area between particles for electrical conductance calculation of particle-to-particle contact. They used two-dimensional (2-D) DEM simulations to investigate the electrical response of particulate systems under mechanical actions. This model was later applied by Descartes et al. [28] in the field of railway engineering to study the impact of the “third-body” layer (e.g., sand) on electrical conduction at wheel-rail interface. Ott et al. [29] utilised three-dimensional (3-D) DEM simulations to explore the calendaring process of granular electrode structures. Hubert et al. [30] characterised the contact area between two overlapping particles as a transmission surface facilitating current flow, and employed DEM simulations to study the variation of resistance in a cylindrical rod subjected to a torsion test. Chapeuil et al. [31] used DEM to model a “third-body” material composed of copper and graphite, and examined the influence of this mixture on the electrical properties of the wheel-rail contact area during train braking. Reynolds et al. [32] calculated the electrical resistance between two overlapping particles from their centres to their common contact surface to evaluate the electrical behaviour of the burden layer in a ferroalloy furnace system through DEM simulations.

The electrical resistance between two overlapping particles involves not only the bulk resistance of each particle but also their contact resistance, and thus requires a comprehensive formulation. From this perspective, Bourbatache et al. [33] proposed an electrical conduction model to investigate electrical transfer in metallic particulate system under mechanical loading. In their model, contact points were defined as circuit nodes, leading to multiple potentials and additional current paths within each particle. Following this, Machado et al. [34] improved the model for application in ball bearings, using DEM simulations to identify defects based on unusual resistance readings. Sangrós Giménez et al. [35] presented a numerical method for modelling electrical resistance networks applicable to all-solid-state batteries. The results showed that DEM simulation results can reproduce experimental data trends and serve as a tool for predicting the conductivity of electrode materials and designing enhanced electrode structures. Yim et al. [36] studied the milling process of ball-milled powder using DEM coupled with experimental observations. The results indicated that the possible path for charge transfer expanded as the coordination number increased, and the electrical resistivity of powders decreased progressively with longer milling times.

This study contributes to the field by presenting an electro-mechanical contact model for current flow from one object to another object through a contact surface, accounting for both the objects' bulk resistance and the contact resistance at their interface [21,37]. The model is verified and validated using previously published analytical and experimental data. In addition, the model is applied to high-pressure torsion (HPT) test to study the effect of materials with different resistivity on the electrical response characteristics at the wheel-rail interface.

2. Description of the electro-mechanical contact model

2.1. Governing equations of particle dynamics

DEM models the behaviour of a particulate system by considering the motion of particles and their state as influenced by their contact with neighbouring particles and the external loads [22,38]. The translational and rotational motions of individual particle are described by Newton's second laws and Euler's law [22,39]:

$$m_i \frac{d\vec{v}_i}{dt} = \sum_{j=1, j \neq i} \left(\vec{F}_{c,ij}^n + \vec{F}_{c,ij}^t \right) + m_i \vec{g} \quad (1)$$

$$I_i \frac{d\vec{\omega}_i}{dt} = \sum_{j=1, j \neq i} \left(\vec{r}_i \times \vec{F}_{c,ij}^t + \vec{M}_{r,i} \right) \quad (2)$$

where $\vec{v}_i$, ω_i , m_i , I_i and $\vec{r}_i$ are the translational velocity, angular velocity, mass, moment of inertia, and radius of particle i , and $\vec{g}$ is the gravitational acceleration. $\vec{F}_{c,ij}^n$ and $\vec{F}_{c,ij}^t$ are the normal and tangential components of the contact force, respectively, exerted by particle j on particle i , and $\vec{M}_{r,i}$ is the rolling friction torque of particle i .

In this study, the force-displacement relationships for two elastic spheres with friction is described using the Hertz-Mindlin (No-Slip) contact model [22,27,40–43], where the components of the contact force in the normal and tangential directions can be written as follows:

$$\vec{F}_{c,ij}^n = \vec{f}_n + \vec{f}_n^d \quad (3)$$

$$\vec{F}_{c,ij}^t = \vec{f}_t + \vec{f}_t^d \quad (4)$$

where f_n and f_t represent the normal and tangential component of contact force, and f_n^d and f_t^d are the normal and tangential damping forces which depend on the overlap between the two particles. Damping plays an important role in allowing the system to dissipate energy and converge to a steady state. The magnitude of the damping coefficient depends on the coefficient of restitution [42]. Therefore, Eqs. (3) and (4) can be written as:

$$\vec{F}_{c,ij}^n = \frac{4}{3} E^* \sqrt{r^*} \delta_n^3 - 2 \sqrt{\frac{5}{6}} \beta \sqrt{k_n m^* \nu_n^{rel}} \quad (5)$$

$$\vec{F}_{c,ij}^t = -k_t \delta_t - 2 \sqrt{\frac{5}{6}} \beta \sqrt{k_t m^* \nu_t^{rel}} \quad (6)$$

where m^* is the equivalent mass, β is the damping coefficient, δ_n and δ_t are the normal and the tangential overlap, ν_n^{rel} and ν_t^{rel} are the normal and tangential components of the relative velocity. In addition, the equivalent Young's modulus (E^*), the effective radius (r^*), the normal spring stiffness (k_n) and the tangential spring stiffness (k_t) are defined as:

$$\frac{1}{E^*} = \frac{(1 - \nu_i^2)}{E_i} + \frac{(1 - \nu_j^2)}{E_j} \quad (7)$$

$$\frac{1}{r^*} = \frac{1}{r_i} + \frac{1}{r_j} \quad (8)$$

$$k_n = 2E^* \sqrt{r^* \delta_n} \quad (9)$$

$$k_t = 8G^* \sqrt{r^* \delta_n} \quad (10)$$

with E_i , ν_i , r_i and E_j , ν_j , r_j defined as the Young's modulus, the Poisson ratio, and radius for particle i and particle j in elastic deformation, respectively. In addition, G^* is the equivalent shear modulus.

2.2. Electrical transfer model

When two solids are compressed together due to an external force, a conductive path is built in their contact region that allows electrons to flow freely across the contact [37]. To evaluate the characteristics of the electrical response for an entire particulate system, one of the key components is to identify the resistance along the conduction path. Therefore, both particle-to-particle and particle-to-wall conduction are considered in this study.

2.2.1. Particle-particle electrical conduction

The electrical resistance from particle i to particle j consists of: the bulk resistance from the centre of particle i to the contact plane (R_i); the contact resistance of the two particles in the contact zone ($R_{c,ij}$); and the bulk resistance from the centre of particle j to the contact plane (R_j), as shown in Fig. 1. The electrical resistance R_{ij} between particle i and particle j can be written as:

$$R_{ij} = R_i + R_{c,ij} + R_j \quad (11)$$

where R_i is the bulk resistance from the centre of particle i to its elastic deformation position, $R_{c,ij}$ is the electrical resistance at the contact area of two particles, and R_j is the bulk resistance of particle j from the elastic deformation position to its centre. In addition, the size of the contact zone between two particles strongly affects their contact resistance, which can be represented by the Holm resistance model as [37,44]:

$$R_{c,ij} = \frac{\rho_i + \rho_j}{4r_c} \quad (12)$$

where ρ_i and ρ_j are the electrical resistivity of particles i and j , respectively; and r_c is the radius of the contact area. According to the classical Hertz theory, when two elastic spherical particles are in contact, the contact radius r_c can be written as [27,39]:

$$r_c = \left(\frac{3f_n r^*}{4E^*} \right)^{\frac{1}{3}} \quad (13)$$

Substituting Eq. (13) into Eq. (12), the contact resistance between two particles can be rewritten as:

$$R_{c,ij} = \frac{1}{4} (\rho_i + \rho_j) \left(\frac{3f_n r^*}{4E^*} \right)^{-\frac{1}{3}} = \frac{1}{2} (\rho_i + \rho_j) \sqrt[3]{\frac{E^*}{6f_n r^*}} \quad (14)$$

For two particles i and j that are in contact with each other, their bulk resistance from the centre of the particles to the contact surface can be written as:

$$R_i = \int_0^{r_i - \delta_i} \rho_i \frac{dz}{\pi(r_i^2 - z^2)} \quad (15a)$$

$$R_j = \int_0^{r_j - \delta_j} \rho_j \frac{dz}{\pi(r_j^2 - z^2)} \quad (15b)$$

where r_i and r_j are the radius and δ_i and δ_j are the deformations of particle i and particle j , during the contact process. In order to determine the amount of elastic deformation for each particle in the contact area, the geometric characteristics of the two particles in contact (as shown in Fig. 1) are taken into account, thus δ_i and δ_j can be calculated as:

$$\delta_i = r_i - D = r_i - \frac{r_i^2 - r_j^2 + L_{ij}^2}{2L_{ij}} \quad (16a)$$

$$\delta_j = r_j - L_{ij} + D = r_j - L_{ij} + \frac{r_i^2 - r_j^2 + L_{ij}^2}{2L_{ij}} \quad (16b)$$

where D is the distance from the centre of particle i to the contact plane, and L_{ij} is the distance between the centres of two overlapping particles. Substituting Eqs. (16a) into (15a) and (16b) into (15b), the bulk resistance of the corresponding half-sphere under elastic deformation for particle i and particle j can be derived as:

$$R_i = \int_0^{\frac{r_i^2 - (r_i - L_{ij})^2}{2L_{ij}}} \rho_i \frac{dz}{\pi(r_i^2 - z^2)} = \frac{\rho_i}{2\pi r_i} \ln \left(\frac{r_j^2 - r_i^2 - L_{ij}^2 + 4r_i L_{ij}}{r_i^2 - r_j^2 + L_{ij}^2} \right) \quad (17a)$$

$$R_j = \int_0^{\frac{r_j^2 - (r_j - L_{ij})^2}{2L_{ij}}} \rho_j \frac{dz}{\pi(r_j^2 - z^2)} = \frac{\rho_j}{2\pi r_j} \ln \left(\frac{r_i^2 - r_j^2 - L_{ij}^2 + 4r_j L_{ij}}{r_j^2 - r_i^2 + L_{ij}^2} \right) \quad (17b)$$

After that, by substituting Eq. (14) and Eqs. (17a) and (17b) into Eq. (11), the electrical resistance from the centre of particle i to the centre of particle j when these two particles are overlapping can be written as:

$$R_{ij} = \frac{\rho_i}{2\pi r_i} \ln \left(\frac{r_j^2 - r_i^2 - L_{ij}^2 + 4r_i L_{ij}}{r_i^2 - r_j^2 + L_{ij}^2} \right) + \frac{1}{2} (\rho_i + \rho_j) \sqrt[3]{\frac{E^*}{6f_n r^*}} + \frac{\rho_j}{2\pi r_j} \ln \left(\frac{r_i^2 - r_j^2 - L_{ij}^2 + 4r_j L_{ij}}{r_j^2 - r_i^2 + L_{ij}^2} \right) \quad (18)$$

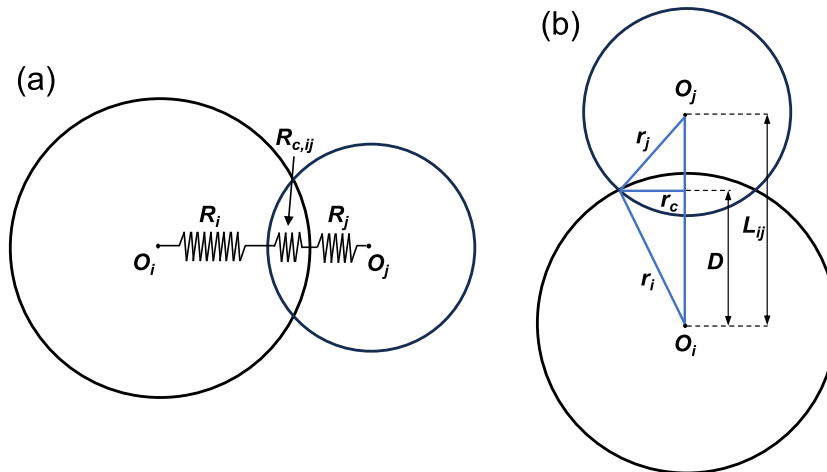


Fig. 1. Schematic of particle-to-particle contact: (a) components of electrical resistance in two particles overlapping, (b) geometry of two particles generating an overlap.

2.2.2. Particle-wall electrical conduction

During the application of mechanical loads to the particulate system, some of the particles come into contact with the boundary walls and generate channels in the overlap region that allow electrons to pass freely, leading to the electrical transfer between the particles and the walls. In this study, the wall is treated as having a uniform and constant electric potential, and when the particle is in contact with the wall, the electrons flow from the wall to the particle through the contact region. Therefore, the electrical resistance R_{iw} resulting from the contact between the particle and the wall should take into account the contact resistance $R_{c,iw}$, and the bulk resistance R_i from the centre of particle i to the contact surface of the wall, as shown in Fig. 2, which can be written as:

$$R_{iw} = R_i + R_{c,iw} \quad (19)$$

The elastic deformation of the particle i subjected to mechanical loading by a wall is δ and the distance from the centre of the particle to the contact surface is $S = r_i - \delta$, which can be used to calculate the bulk resistance of particle i and written as follows:

$$R_i = \int_0^S \rho_i \frac{dz}{\pi(r_i^2 - z^2)} = \frac{\rho_i}{2\pi r_i} \ln \left(\frac{2r_i - \delta}{\delta} \right) \quad (20)$$

At the overlapping area of the particle and the wall, the contact radius of the overlapping circular surface is determined by Hertz contact theory [27,39] and Holm resistance model [37,44] can then be employed to obtain the contact resistance:

$$R_{c,iw} = \frac{1}{4}(\rho_i + \rho_w) \left(\frac{3f_n r^*}{4E^*} \right)^{-\frac{1}{3}} = \frac{1}{2}(\rho_i + \rho_w) \sqrt[3]{\frac{E^*}{6f_n r_i}} \quad (21)$$

where ρ_w is the electrical resistivity of the wall. Substituting the Eq. (20) and Eq. (21) into Eq. (19), the electrical resistance between particle and wall can be written as:

$$R_{iw} = \frac{\rho_i}{2\pi r_i} \ln \left(\frac{2r_i}{\delta} - 1 \right) + \frac{1}{2}(\rho_i + \rho_w) \sqrt[3]{\frac{E^*}{6f_n r_i}} \quad (22)$$

2.3. Electro-mechanical coupling and assumptions

Generally, the intrinsic electrical properties of the material (i.e., electrical resistivity), the size and number of particles, the roughness and contamination film on the particle surface, in addition to the temperature variations in the particulate system affect the properties of electrical conduction [33]. In this study, it has been assumed that the temperature of the particulate system is constant, the surfaces of the particles in the system are smooth, and the effect of contamination films

is negligible.

An example of a simple particulate system is shown in Fig. 3a, the particles are packed into a rectangular container where the top plate ad and bottom plate bc are electrically conductive, while the sides ab and dc are electrically insulated. The entire system is connected to a direct current (DC) and a normal load is imposed on ad . After ad comes into contact with the particles, current I_0 flows from ad into the particles and across the particulate system to bc . By referring to the network model initially proposed for thermal conduction [45,46] and the nodal analysis method [47,48], the whole particulate system can be regarded as a network of circuits. Each particle within the granular system can be regarded as a node and the path between the centres of two adjacent particles is considered a branch with associated resistance R_{ij} . In addition, drawing from the principles of circuit analysis [49], the two conductive boundaries ad and bc can be regarded as node 0 and node 1 respectively. Consequently, the conductive path generated by the particle and the wall during contact is considered as a branch, incorporating an appropriate resistance denoted R_{iw} . As illustrated in Fig. 3a, the relationship between the electrical properties in the particulate system can be represented as the following form:

$$\underbrace{\begin{pmatrix} G_{00} & G_{10} & G_{i0} & \dots & G_{n0} \\ G_{01} & G_{11} & G_{i1} & \dots & G_{n1} \\ G_{0i} & G_{1i} & G_{ii} & \dots & G_{ni} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ G_{0n} & G_{1n} & G_{in} & \dots & G_{nn} \end{pmatrix}}_{\mathbf{G}} \underbrace{\begin{pmatrix} V_{top} \\ V_{bottom} \\ V_i \\ \vdots \\ V_n \end{pmatrix}}_{\vec{V}} = \underbrace{\begin{pmatrix} -I_0 \\ I_0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\vec{I}} \quad (23)$$

where $\mathbf{G}$ is the conductivity matrix, $\vec{V}$ and $\vec{I}$ are the vector of electric potential and current, respectively. Based on the features of the parallel circuit, the sum of the branch currents I_{iw} on each particle in contact with ad is equal to the total current I_0 entering/leaving the boundary $\partial\Omega$, as follows:

$$\sum_i^{N_w} I_{iw} = I_0 \quad i \in \partial\Omega \quad (24)$$

where N_w is the number of particles in contact with the wall. In Fig. 3b, after the branch current enters particle i from ad , it continuously flows through particle j and k . Following Ohm's law, Kirchhoff's current law, and the relationship between mechanical loading and current [33], the above circuit properties can be expressed as:

$$I_{iw} = \frac{F_n}{F_{ext}} I_0 \quad (25)$$

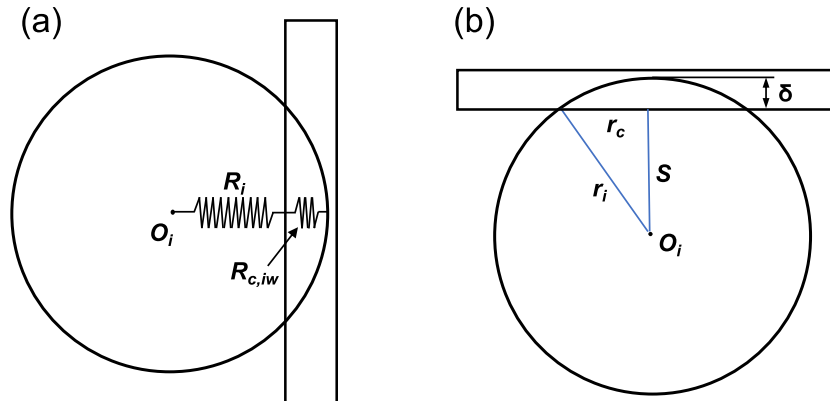


Fig. 2. Schematic of particle-to-wall contact: (a) components of electrical resistance when a wall and a particle are overlapping, (b) geometrical characteristics of a particle and a wall in their contact region.

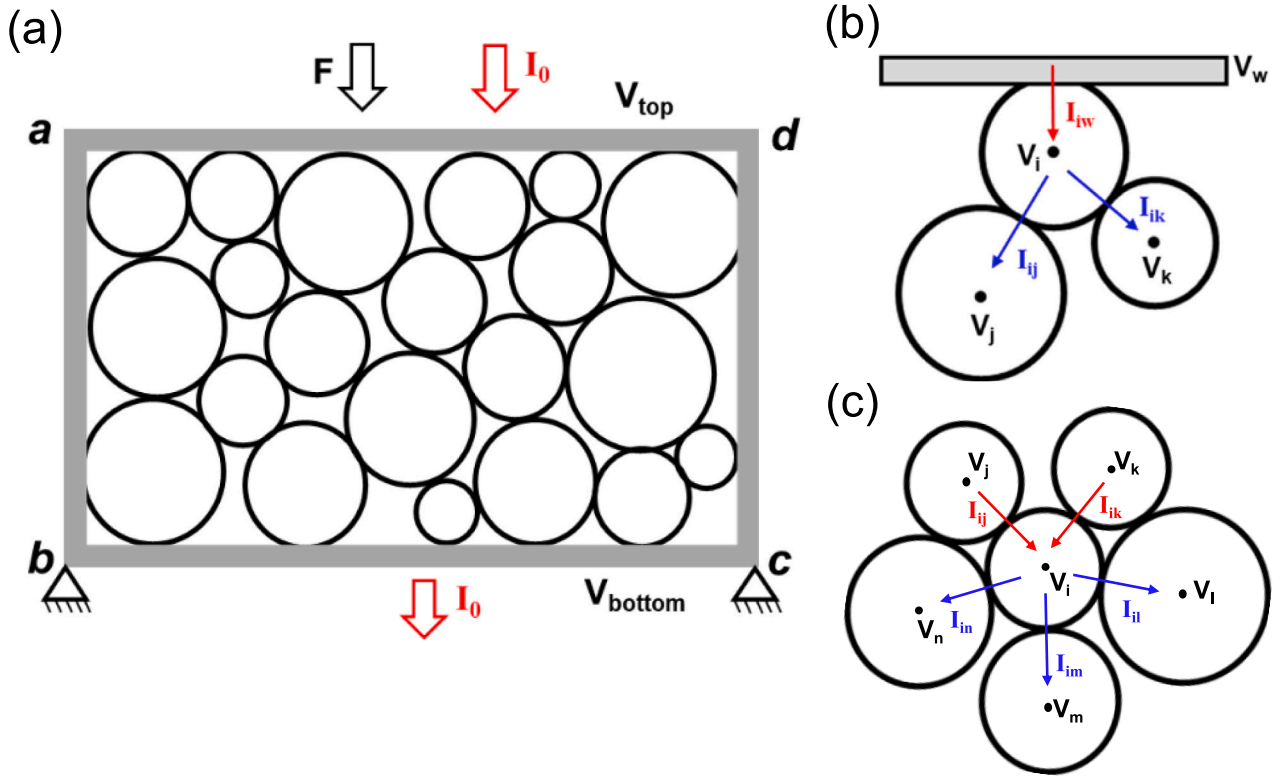


Fig. 3. (a) Schematic of a setup for characterising the electrical transfer properties of a particulate system under mechanical loading, (b) Schematic of the electrical conduction when particle i is in contact with neighbouring particles and the wall (The red arrows indicate the current entering the particle and the blue arrows indicate the current exiting the particle), (c) Schematic of a granular assembly showing the electrical conduction of particle i in contact with neighbouring particles. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$\sum_j^{N_{nei}} I_{ij} + I_{iw} = \sum_j^{N_{nei}} \frac{(V_i - V_j)}{R_{ij}} + \frac{(V_w - V_i)}{R_{iw}} = \sum_j^{N_{nei}} C_{ij} (V_i - V_j) + C_{iw} (V_w - V_i) = 0 \quad (26)$$

where N_{nei} is the number of particles adjacent to particle i , V_w , V_i , and V_j are the electric potential of top plate ad , particle i and particle j , and R_{ij} is the resistance between these two particles which is the inverse of the conductance C_{ij} , R_{iw} and C_{iw} are the resistance and conductance, respectively, between the top plate ad and particle i .

Fig. 3c shows a simple granular assembly in which particle i is in contact with surrounding particles. Taking particle i and particle j as an example, a linear relationship of current I_{ij} and potential drop between these two particles can be created using Ohm's law, where the inter-particle electrical conductance C_{ij} can be considered as a proportionality constant. According to Kirchhoff's current law, the sum of the currents entering and leaving particle i should be zero, which can be written as:

$$\sum_j^{N_{nei}} I_{ij} = \sum_j^{N_{nei}} \frac{(V_i - V_j)}{R_{ij}} = \sum_j^{N_{nei}} C_{ij} (V_i - V_j) = 0 \quad (27)$$

As illustrated in Fig. 4, the procedure of the electro-mechanical coupling is summarised as follows:

- (1) Generate and initialise the particulate system;
- (2) Determine the positions and velocities for each particle;
- (3) Calculate the contact radius via Eq. (13) and update the geometry transformations of particles during mechanical loadings;
- (4) Compute the electrical resistance for both particle-to-particle and particle-to-wall via Eq. (18) and Eq. (22) according to the contact information;

- (5) Evaluate the local electrical conduction properties via Eq. (26) and Eq. (27);
- (6) Predict the global electrical network of the entire particulate system via Eq. (23);
- (7) Update the electric potential of each particle within the assembly and store the data of this time step; and.
- (8) Return to step (2) to repeat the entire electro-mechanical DEM simulation for the next timestep until reaching the end time.

3. Validation of the numerical model

The above numerical scheme is implemented in the DEM commercial software package EDEM™ [50] through the application programming interface (API) using C++ programming language. The validity of the implementation is examined using single-particle compression, particle-particle contact, chains of particles, and particle bed tests. The numerical results are compared against published analytical solutions and experimental data. In addition, to balance the stability of the numerical calculations with the computational cost and to achieve a reasonable simulated phenomena within an acceptable computation time, a 20% of Rayleigh's timestep [51,52] is utilised for all the cases in this study.

3.1. Comparison with the analytical solutions

In this section, a single-particle compression test and a particle-particle contact test are reported. The parameters used in the simulations are listed in Table 1, which are taken from [18].

3.1.1. Case1: Single-particle compression test

To evaluate the solution proposed in this study for calculating the electrical resistance of the particle-to-wall system, the overall electrical resistance R_{wiw} of the particle in contact with two walls is derived on the

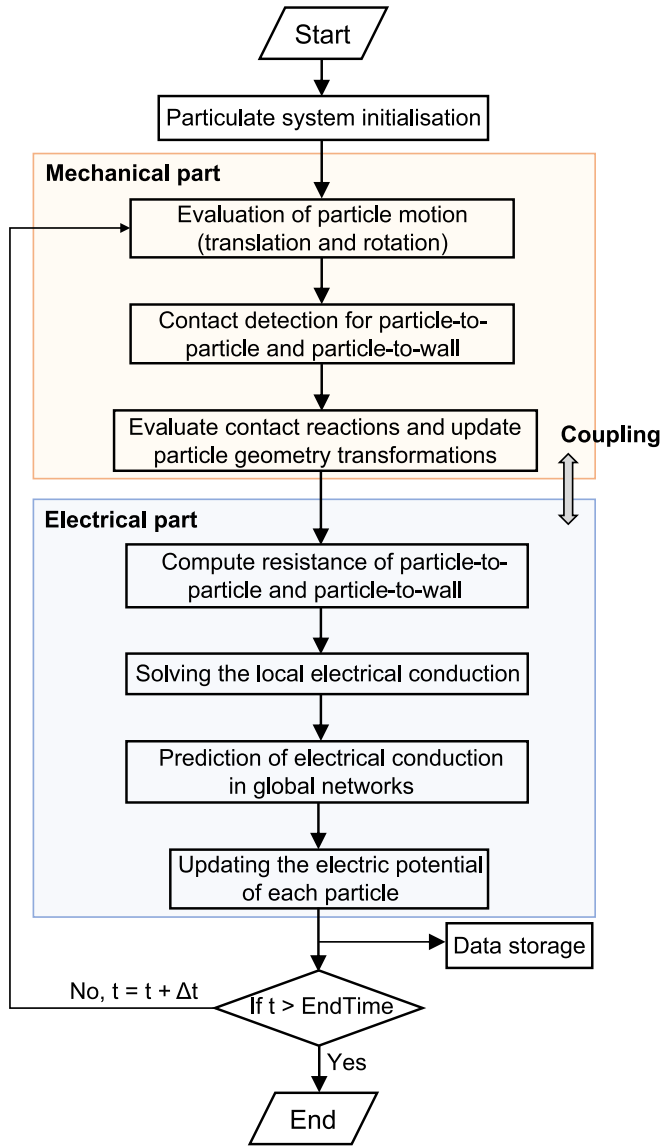


Fig. 4. Flowchart of the solution in the DEM framework for the electro-mechanical coupled model.

basis of Eq. (22):

$$R_{wiw} = \frac{\rho_i}{\pi r_i} \ln \left(\frac{2r_i}{\delta} - 1 \right) + (\rho_i + \rho_w) \sqrt[3]{\frac{E^*}{6f_n r_i}} \quad (28)$$

A single-particle compression test is modelled using copper pads and a nickel ball to investigate the variation of system resistance under different mechanical loading conditions. The simulation results are compared with the analytical solutions from Yim and Paik [18]. In Fig. 5, it can be observed that the numerical results almost perfectly match with the analytical results. The electrical resistance of the particle-to-wall contact system decreases with increasing mechanical force. It can be noticed that the electrical resistance of the system experiences a significant decrease as the applied mechanical force increases from 0 N to 10 N. This is due to the very small contact surface between particle and wall under very low mechanical force limits electron flows, leading to an excessively high electrical resistance. As the mechanical force continues to increase, the change in contact area gradually becomes less significant. Therefore, the electrical resistance of this system does not show significant variations even as the mechanical force reaches a high level.

Table 1

Adopted parameters of the materials in the DEM simulation.

Parameters	Unit	Value
Nickel ball		
Poisson's ratio	–	0.312
Particle density	kg/m ³	8908
Young's modulus	GPa	199.5
Particle diameter	mm	1.0
Electrical resistivity	Ω m	8.0×10^{-8}
Copper pad		
Poisson's ratio	–	0.343
Density	kg/m ³	8960
Young's modulus	GPa	129.8
Electrical resistivity	Ω m	1.8×10^{-8}

3.1.2. Case2: Particle-particle contact test

As shown in Fig. 6, the numerical results of electrical resistance for a particle-particle contact system under different mechanical forces are compared with two published analytical solutions [29,49]. The numerical results are in excellent agreement with the analytical solutions from Birkholz et al. [49], and Ott. et al. [29]. A similar phenomenon to the single-particle compression test is observed, i.e., the electrical resistance of the particle-to-particle system decreases sharply as the mechanical force increases from a very small value. After that, the effect of the increase in mechanical force on the resistance of the particle-to-particle system is very small.

3.2. Comparison with experimental results

3.2.1. Chain of stainless-steel particles

A one-dimensional (1-D) chain of stainless-steel particles is modelled to investigate the effect of sequential contacts between particles on overall electrical conduction [12,13,15]. The schematic of the experimental setup is shown in Fig. 7, in which a chain of N ($N_{\max} = 50$) identical stainless-steel particles is assembled via a series connection and placed inside a cylindrical container composed of polyvinylchloride (PVC) material. A DC current is then supplied to this particle-chain system while a piston, with a force range of 1 to 500 N, applies a static force to the chain. When the particles come into contact with each other, a series circuit is generated, which can be utilised to calculate the resistance for the particle-chain system.

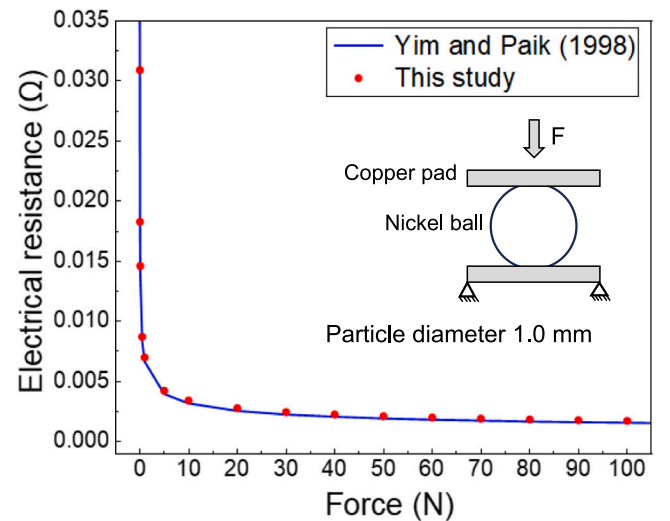


Fig. 5. Evolution of electrical resistance for particle-to-wall contact under different mechanical forces (red dots are simulation results, blue line is analytical results, and the inset figure is a schematic diagram of a single-particle compression setup). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

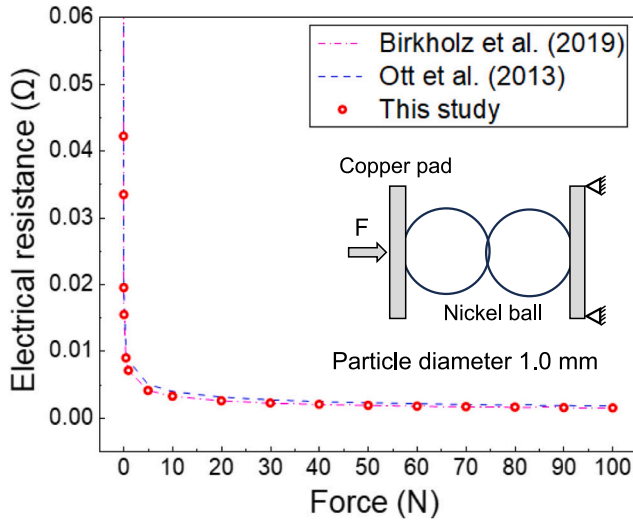


Fig. 6. Evolution of electrical resistance for particle-to-particle contact under different mechanical forces (red dots are simulation results, blue and pink dash lines are analytical results, and the inset figure is a schematic diagram of a particle-particle contact setup). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Three different DC current intensities are applied to the chain of stainless-steel particles to study how its electrical conduction varies under the influence of mechanical forces. The material parameters used in the simulation are listed in Table 2, which are taken from Falcon et al. [12].

Fig. 8 illustrates an example of the electric potential distribution in a chain of stainless-steel particles when subjected to a static force of 100 N and a current of 0.0001 A. It can be observed that the particle chain exhibits a potential drop from the left side to the right side. To be specific, electrons will flow through the particle chain from the anode (left side) to the cathode (right side) in this system. During the compression process, the particles will be slightly deformed under the action of static force, and the contact area between the particles will increase, thus affecting the bulk resistance of particles and the contact resistance.

Fig. 9 compares the experimental data and DEM results for the chain of stainless-steel particles under different static forces for three specified current intensities. The electrical resistance of the particle-chain system gradually decreases with increasing static force. For the initial stage of static force application, the DEM results underestimate the electrical resistance of the particle-chain system. The small static force is insufficient to break the oxide layers on the surface of the particles, hindering electron penetration and the formation of a conductive channel. This results in the inclusion of the resistance of oxide layer in the measured experimental data. A closer match with the experimental results is achieved when the static force exceeds 100 N, aligning with observations from previous theoretical interpretations [12,13,53].

3.2.2. Particle bed test

The 3-D spherical steel particle bed test described by [33] is used to

examine the validity of the proposed numerical scheme. In this case, the electrical conduction of the granular system is affected by not only the nature of the particle materials and contacts, but also the structural characteristics of the entire granular network. Fig. 10 shows a schematic diagram of a compaction experiment on a particle bed in which steel particles are placed in a cylindrical container with a diameter of 5 cm. To simplify the simulation, the top and bottom plates are assumed to be made of the same material as the particles, and the cylinder wall is electrically insulated. The anode and cathode of the power supply are connected to the top and bottom plates of the cylindrical container, respectively.

A current of 0.01 A is applied as a boundary condition to the top and bottom plates; simultaneously, a voltage of 0 V is applied to the bottom plate, thus specifying that the current flows from the top plate to the bottom plate. At the beginning of the numerical experiment, a compressive load is applied to the top plate. When the top plate is in contact with the steel particles, the DC current enters the particle bed from the top plate and then transfers to the bottom plate. The electric potential drop is obtained between the top and bottom sides of the granular system once it stabilises, enabling the calculation of the electrical resistance of the particle bed. The properties of steel particles and simulation parameters are listed in Table 3, which are taken from [33].

Fig. 11 shows the evolution of electrical conduction for a spherical steel particle bed under simultaneous application of an electric current and a compressive load. As the compressive load continues to be applied, a noticeable potential gradient appears and propagates throughout the entire particle bed. When the loading time reaches 0.5 s, the potential distribution of the granular system is in a steady state (Fig. 11a), which is consistent with the observations of previous DEM studies [30,33].

Fig. 12 compares the 3-D numerical predictions and experimental data for particle beds with different numbers of layers; previous 2-D DEM simulation results are also plotted for comparison [33]. Note that the steel particles used in the experiments are slightly oxidised, resulting in significantly smaller observed than calculated electrical resistance at lower mechanical loads (i.e. for forces <50 N in Fig. 12). As the mechanical load increases to the point where the oxidised layer can be crushed and the surface of the particles exposed, agreements between 3-D numerical predictions and experimental results are achieved.

In Fig. 12a, a compressive load and DC current are applied to a 3-layer particle bed. The 2-D and 3-D models both predict decreasing resistance as the mechanical force increases, with small differences in the form the relationship, but both in reasonable agreement with the

Table 2

Model parameters used for 1-D stainless-steel bead chain test.

Parameters	Unit	Value
Number of particles	–	41
Particle diameter	mm	8
Poisson's ratio	–	0.27
Particle density	kg/m ³	7750
Young's modulus	Pa	1.95×10^{11}
Electrical resistivity	$\mu\Omega$ cm	72
Imposed current intensity	A	0.01, 0.001, 0.0001

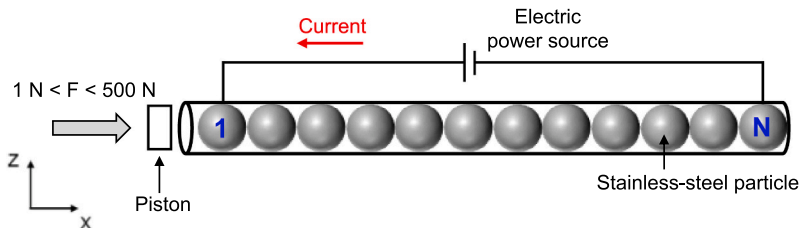


Fig. 7. Schematic of the experimental setup for a chain of stainless-steel particles.

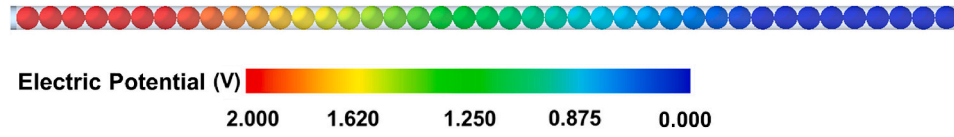


Fig. 8. The electric potential distribution of the stainless-steel particle chain in the DEM simulation with static force of 100 N and current of 0.0001 A.

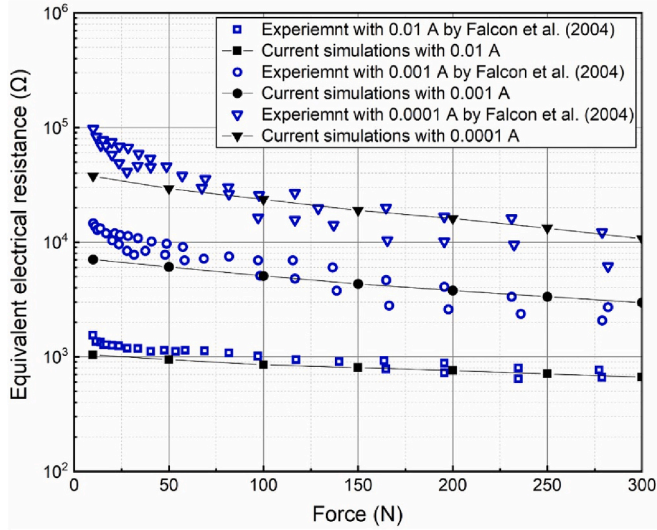


Fig. 9. Comparison between numerical predictions and experimental data for equivalent electrical resistance of stainless-steel particle-chain system under various mechanical forces.

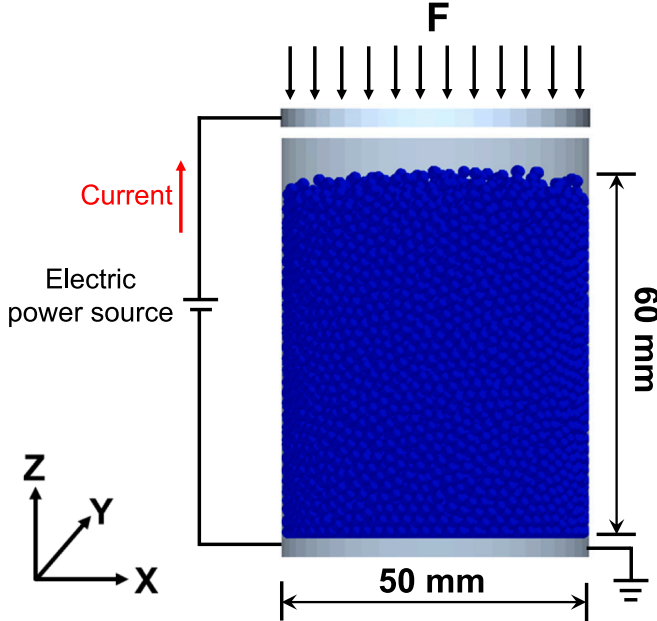


Fig. 10. Schematic of a compaction experiment on the particle bed subjected to an imposed DC current.

experimental data. However, as the number of particle bed layers increases the difference between the 2-D and 3-D numerical results increases (Fig. 12b and c). The 2-D simulation, lacking a third dimension, fails to capture the complexity of particle interactions within the granular system. This shortcoming constrains the model's ability to accurately capture the electrical conduction for the particulate network, which leads to an apparent and increasing divergence of the 2-D

Table 3

Material properties and numerical parameters for the steel particle bed test.

Parameters	Unit	Value
Particle diameter	mm	2
Poisson's ratio	–	0.3
Particle density	kg/m ³	7850
Young's modulus	Pa	2×10^{11}
Electrical resistivity	Ω m	1.71×10^{-8}
Imposed current	A	0.01

calculations from the experimental results.

4. Application: High pressure torsion test

In the following, the electro-mechanical contact model is examined against the HPT test, a laboratory technique that can be utilised to assess the electrical response at the wheel-rail interface [17]. This case study aims to predict the effect of materials with different conductivities on the electrical transfer between wheel and rail under over-application of third body traction enhancing particles.

4.1. Model setup

The real simulation model of the HPT setup is shown in Fig. 13. A geometry bin is created in the shape of a cylindrical shell with an inner diameter of 10.5 mm, an outer diameter of 18 mm, and a height of 2 mm, which allows the material to be placed in the wheel-rail contact region. In this case study, the amount of each type of material is managed by maintaining a consistent volume fraction, which ensures that no direct contact is made between the wheel and rail specimens.

Subsequently, a 0.5 V_{DC} power supply is connected to the HPT system while applying a vertical load to the wheel specimen, as shown in Fig. 13. The value of 0.5 V_{DC} represents the worst-performing low-voltage DC track circuit in UK rail operations, 79% of wrong side track circuit failures occurred on DC circuits between 2012 and 2022 [54]. The wheel specimen crushes the material into fragment layers as it moves closer to the rail specimen; current is transmitted from the wheel specimen to the rail specimen via the fragment layers creating a closed circuit. When the particles are completely crushed and the wheel specimen is not further displaced, a uniform angular velocity is applied to the rail specimen. Voltage and current data collected during the rotation can be used to calculate the electrical resistance of the HPT system.

Two newly-developed commercial conductive materials (for confidentiality, these two materials are named Product B and Product D) and silica sand are employed in this case study. Particle characterisation of these materials is reported in [17]. As the HPT data are primarily collected during the rotational stage when the particles are already crushed, particle shape does not play a substantial role in the result. Therefore, the shape of particle for each material is modelled as a sphere. To generate a clumped particle, the crushed samples collected from railhead are analysed using laser diffraction and spherical elements representing the fragment size distribution are generated using an in-house Python code. In this study, the bonded particle model (BPM) [55] is employed to simulate the behaviour of the three crushable materials, as illustrated in Fig. 14. Each particle is represented as a cluster of independent spherical fragments bonded together with finite-sized bonds. The translational and rotational motions of each fragment are

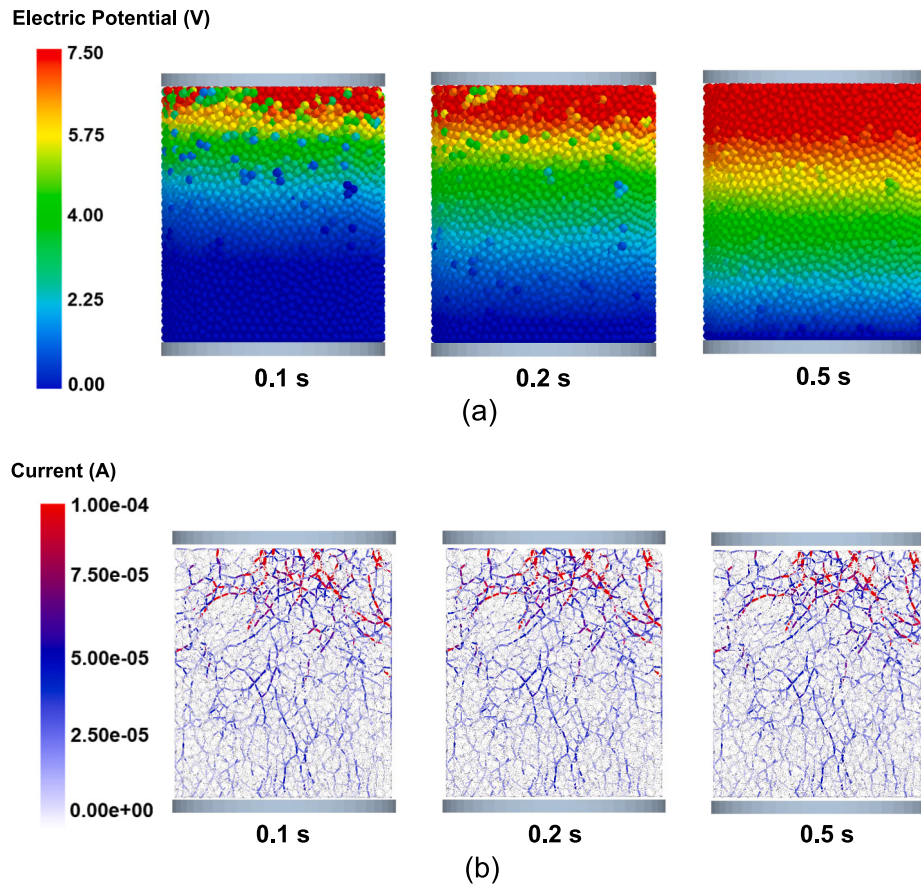


Fig. 11. Snapshots of the evolution of the electrical conduction for a steel particle bed at different times when it is subjected to a mechanical force of 100 N and a current of 0.01 A: (a) the electric potential distribution of particle bed and (b) the current intensity distribution of particle bed.

communicated to other bonded fragments through these bonds under the compression of the wheel specimen. As the wheel specimen continues to compress the particles, the normal and tangential stresses acting on the bond will exceed its specified critical strength value, leading to bond fracture and particle breakage. The particles are completely crushed during high pressure torsion testing, for more details readers are referred to our previous work [56].

The numerical parameters used for each material, wheel specimen, rail specimen, and the operation of the HPT setup are listed in Table 4, which are based on [17,56,57].

4.2. HPT simulation results

Fig. 15 shows the electrical conduction of the fragment layers formed by the breakage of the three different materials at the end of the HPT simulation. The electric potential of the fragment layers of each material decreases from top to bottom. The current is transferred through the contact surfaces between the fragments and the amount of current being transmitted is proportional to the contact area. In addition, an analogy can be identified between the current paths and the chains of force within the granular system, which is in agreement with the findings of Bourbatache et al. and Machado et al. [33,34].

Fig. 16 illustrates that the high electrical conductivity of Products B and D ensures that the HPT system maintains low resistance values ($<10 \Omega$), even with the application of excessive amounts of these two conductive materials. However, an excess of Silica sand leads to significant resistance, the fragment layers produced by sands prevent metal-to-metal contact between wheel and rail specimens while the Silica sand particles have extremely weak conductivity and therefore the current intensity is extremely low, as seen in Fig. 16c.

At the start of the HPT simulation, the HPT system is an open circuit, resulting in an infinite value of the electrical resistance as the wheel specimen is not in contact with the particles. At $t = 0.005$ s, the electrical resistance drops to a finite measurable value, the crushed particles build conduction paths for the wheel and rail specimens that allows the flow of current. Later, as the rail specimen starts to rotate, the electrical resistance of the HPT system fluctuates significantly. At this stage, the fragments transform from static (being crushed) to dynamic (rotating), and the contact conditions and local alignment between the fragments change, leading to fluctuation in the contact forces, and consequently the current. The HPT system is completely sliding at around 0.02 s, the contact state between fragments is stabilised, and its resistance value is gradually smoothed.

Fig. 17 compares the experimental data and the HPT numerical results for over-application of each material at the wheel-rail contact region. The numerical predictions for each material are within the boundaries of the experimental data from [17]. There is some stochasticity in the experimental results, attributed to heterogeneity of the material in each repetition of the HPT test. Moreover, in Fig. 17c, certain experimental resistance data (1–100 Ω) appear considerably smaller than the numerical results. This discrepancy may be attributed to the silica sand particles fragmenting and being expelled from the wheel-rail interface under compressive force. Alternatively, the fragments may be displaced during rotation, leading to direct metal-to-metal contact between the wheel and rail specimens.

Conductive materials B and D are selected to evaluate the influence of the fragments generated after particle crushing on the electrical transfer at the wheel-rail specimen interface at the microscopic scale. Fig. 18a shows the number of fragments that are electrically active for Product B and D at the wheel-rail interface. The number of electrically

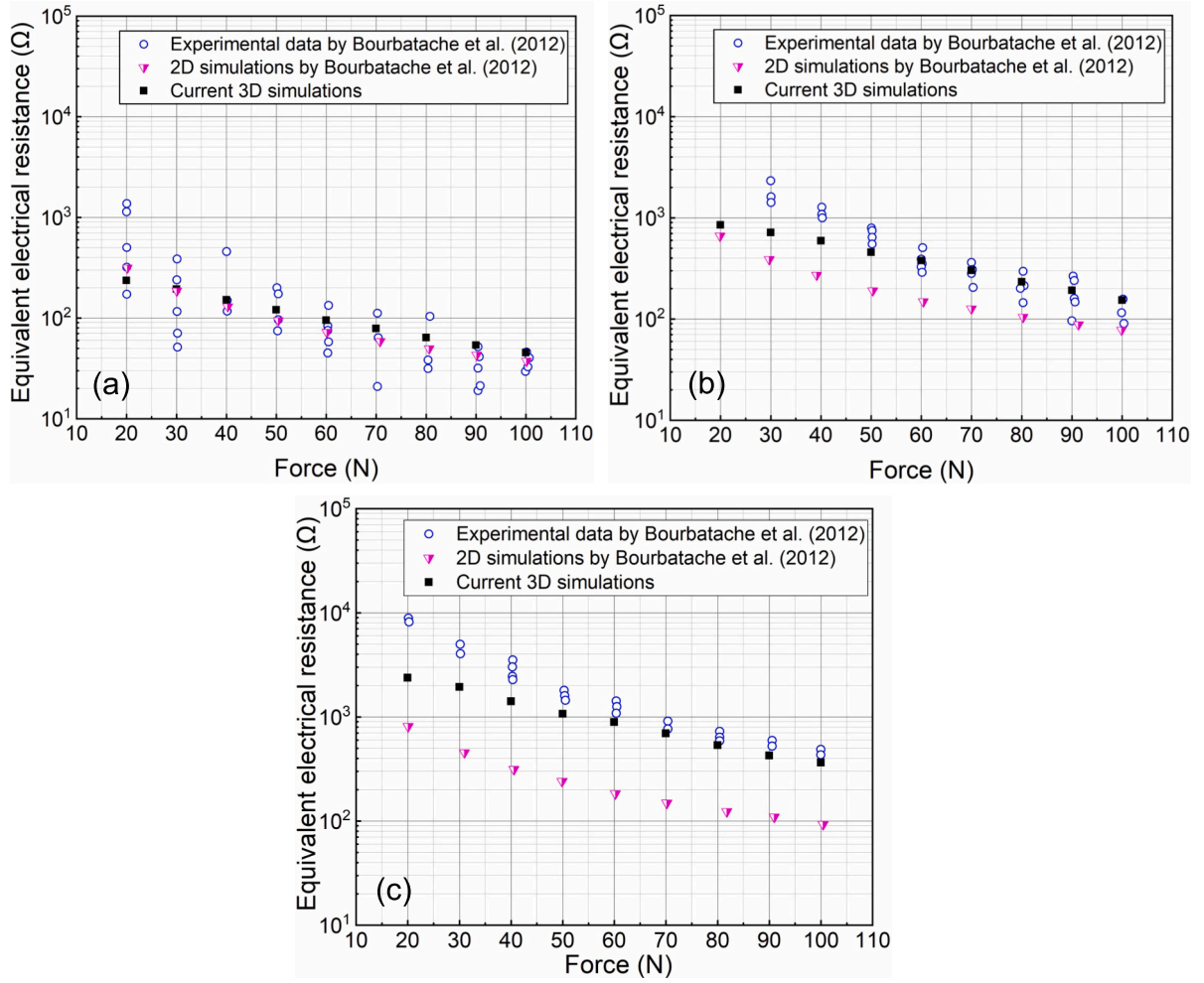


Fig. 12. Variation of electrical resistance under different mechanical forces in comparison with experimental data and 2-D simulation results: (a) 3-layer particle bed, (b) 5-layer particle bed, and (c) 10-layer particle bed.

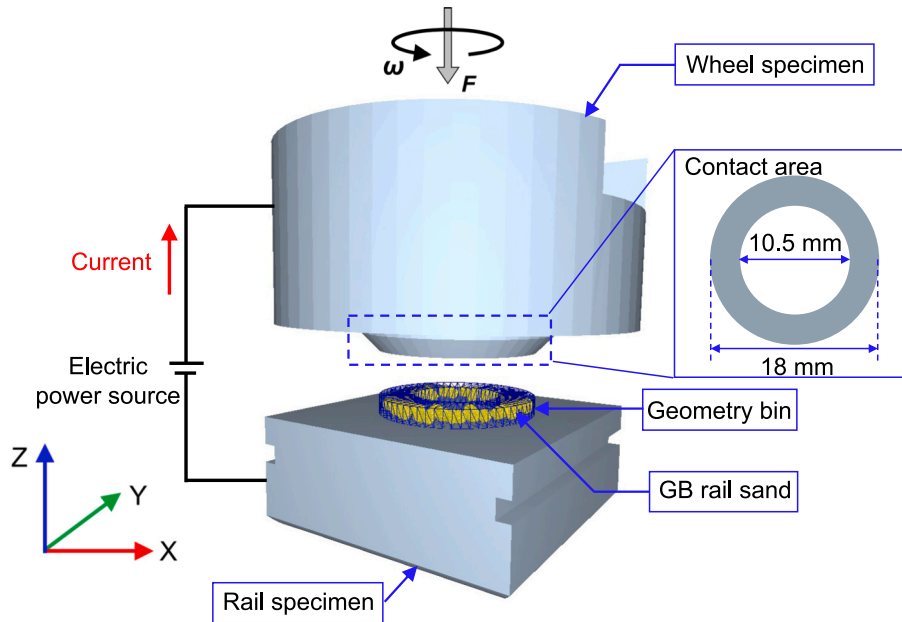


Fig. 13. The HPT setup connected to a DC circuit and the geometrical parameters of its contact area between wheel and rail specimen.

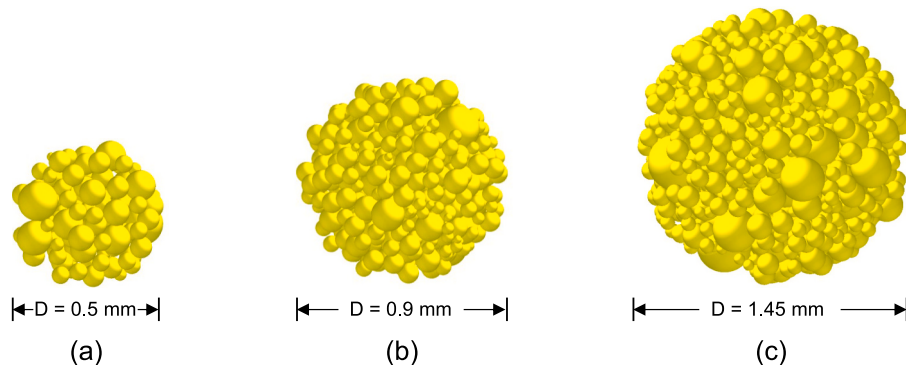


Fig. 14. Particle modelling of three different crushable materials: (a) Product B with particle size 0.5 mm, (b) Product D with particle size 0.9 mm, and (c) Silica sand with particle size 1.45 mm.

Table 4

Properties of materials and specimens and related parameters for the HPT test.

Parameters	Unit	Value
Product B		
Particle diameter (d_p)	mm	0.5
Number of particles (N_p)	–	1051
Fragment diameter (d_f)	mm	0.06–0.2
Number of fragments (N_f)	–	154,497
Poisson ratio (ν)	–	0.3
Density (ρ)	kg/m ³	3.81×10^4
Young's modulus (E)	Pa	7×10^8
Electrical resistivity (ρ_e)	Ωm	4.31×10^{-5}
Product D		
Particle diameter (d_p)	mm	0.9
Number of particles (N_p)	–	180
Fragment diameter (d_f)	mm	0.06–0.3
Number of fragments (N_f)	–	82,440
Poisson ratio (ν)	–	0.3
Density (ρ)	kg/m ³	3.75×10^4
Young's modulus (E)	Pa	7×10^8
Electrical resistivity (ρ_e)	Ωm	4.31×10^{-5}
Silica sand		
Particle diameter (d_p)	mm	1.45
Number of particles (N_p)	–	43
Fragment diameter (d_f)	mm	0.06–0.38
Number of fragments (N_f)	–	55,384
Poisson ratio (ν)	–	0.3
Density (ρ)	kg/m ³	2.65×10^4
Young's modulus (E)	Pa	7×10^8
Electrical resistivity (ρ_e)	Ωm	5.56×10^6
Wheel specimen		
Poisson ratio (ν)	–	0.28
Density (ρ)	kg/m ³	7850
Young's modulus (E)	Pa	2.3×10^{11}
Electrical resistivity (ρ_e)	Ωm	1.43×10^{-7}
Normal pressure	MPa	6
Rail specimen		
Poisson ratio (ν)	–	0.3
Density (ρ)	kg/m ³	7850
Young's modulus (E)	Pa	2.1×10^{11}
Electrical resistivity (ρ_e)	Ωm	1.43×10^{-7}
Angular velocity	deg/s	1
Bond properties		
Normal stiffness (k_n)	N/m ³	1×10^{10}
Shear stiffness (k_s)	N/m ³	1×10^{10}
Critical normal stress (σ_{max})	Pa	1×10^8
Critical shear stress (τ_{max})	Pa	1×10^8
Interaction properties		
Restitution coefficient (e)	–	0.8
Friction coefficient (μ)	–	0.5

active fragments is most concentrated in the range of smaller fragment sizes. However, this does not indicate that smaller fragments are more electrically active or contribute more to electrical transfer in the particulate system than larger ones. To address this, an electroactivity index

is proposed in this study for evaluating the contribution of each fragment sieve cut on the bulk electrical conduction behaviour.

$$\text{Electroactivity Index} = \frac{\text{Number of electroactive fragments at each size}}{\text{Total number of fragments at each size}} \quad (29)$$

Fig. 18b shows the degree of electrical activity for each size of fragment as evaluated by the electroactivity index. Overall, the degree of electroactivity of the fragment increases with size. This suggests that a portion of the small-sized fragments fail to make effective contact with the wheel-rail specimen or with other fragments, limiting their participation in conductive paths in the particulate system. In contrast, large-sized fragments are widely involved in the force-chain network, leading to current flow through them. This observation aligns with the understanding that current paths in particulate systems follow the force-chain network [33,34]. Additionally, the lack of current flow through some non-contacting small fragments results in a zero electric potential for these fragments, consistent with observations in Fig. 15.

5. Discussion

The electro-mechanical model proposed in this study, for electrical resistance of the particle-to-particle and particle-to-wall cases, takes into account the contact resistance of the overlapping region between two solids together with the bulk resistance of the particles. This modification enables the application of the model to particulate systems with different particle size distributions. However, the proposed model still has some limitations due to the assumptions made. These are discussed below, along with suggestions for future work.

The numerical results of the validation section show a similar trend in the electrical resistance of metal particle systems to the elastic Hertz theory (i.e., $R \sim F^{1/3}$) under mechanical loading conditions. This suggests that contact resistance plays a dominant role. As the model has not yet considered the effect of a contamination layer on the particle surface, there can be an apparent gap between the numerical predictions and experimental data at lower static forces. To better investigate how the contamination layer blocks metal-to-metal conductive paths on a microscopic scale, further model development is necessary. This can also help understand the resistance from the contamination layer as a barrier to electrical transmission. It is also noted that the real contact area between particles may be larger than the nominal contact area due to the surface roughness of real particles [59], and therefore more advanced models need to be developed in the future to track the current transfer at the contact interface between rough particles.

The present numerical framework is based on the Hertz-Mindlin model and is therefore suitable for modelling the electrical conduction behaviour of particulate systems under dry contact conditions. However, real-world scenarios are more complex. For example, applying this electro-mechanical model to actual railway projects requires

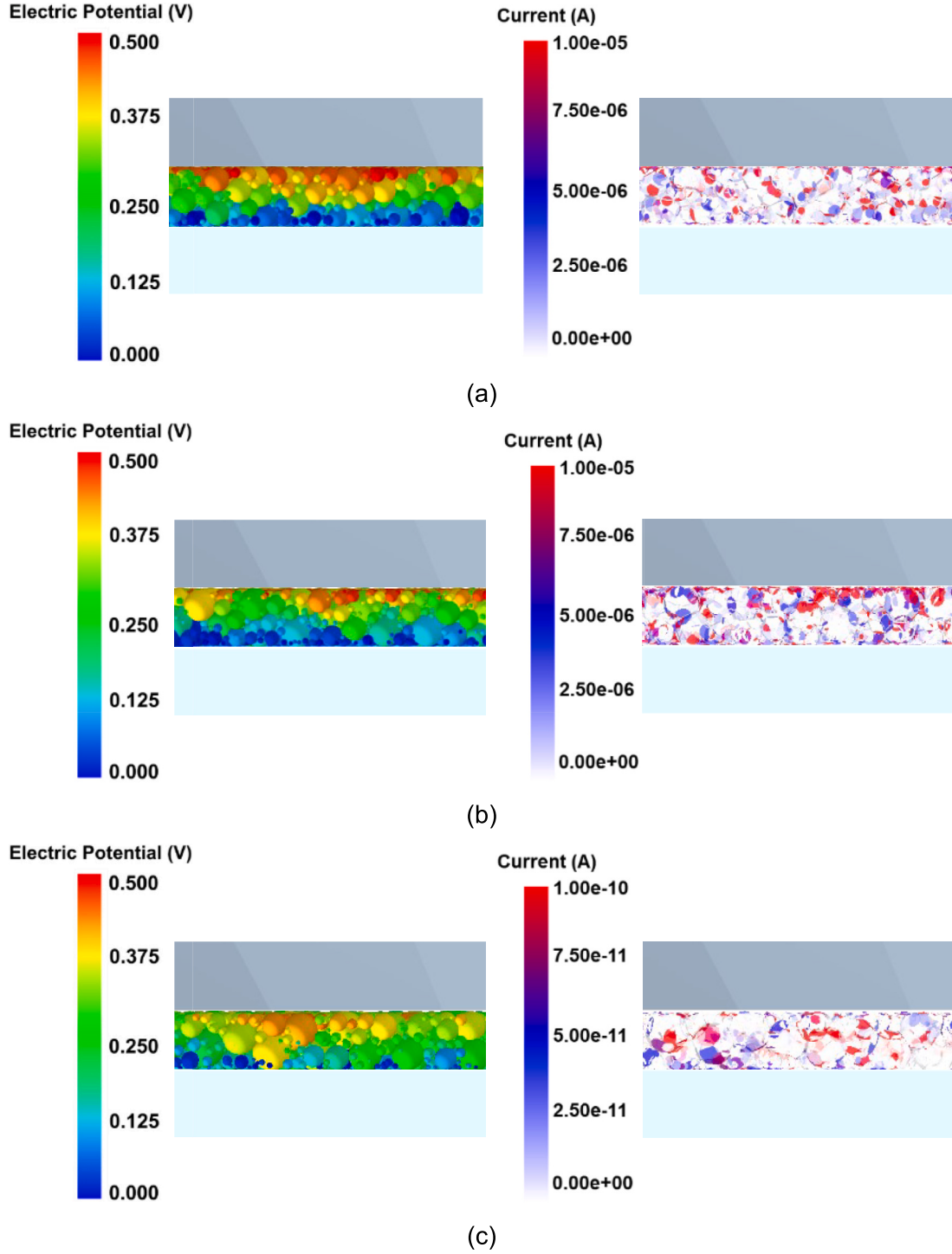


Fig. 15. Snapshots of the electric potential distribution and current intensity distribution across the fragment layers formed by the crushing of the three different materials: (a) Product B, (b) Product D, and (c) Silica sand.

considering various contact conditions at the wheel-rail interface, such as wet [60] or leaf contaminated [61] conditions. Consequently, the proposed model should be refined in future work to align with experimental and field data. In addition, this model is primarily based on elastic particle-to-particle and particle-to-wall contacts, although it is applicable to DEM simulations of many fundamental particulate systems, further model improvements are desirable, for example, wheel-rail contact involving elasto-plastic deformation [62].

Despite the limitations mentioned above, the proposed formulation provides a new route for the analysis of the electrical response of

particulate systems under mechanical loadings. Its potential applications extend far beyond the case study outlined here, such as the compression process of metal powders, drug delivery systems and energy storage.

6. Conclusions

This paper presents an electro-mechanical contact model to investigate the electrical response of the particulate systems under mechanical loading conditions. The model was verified through analytical solutions

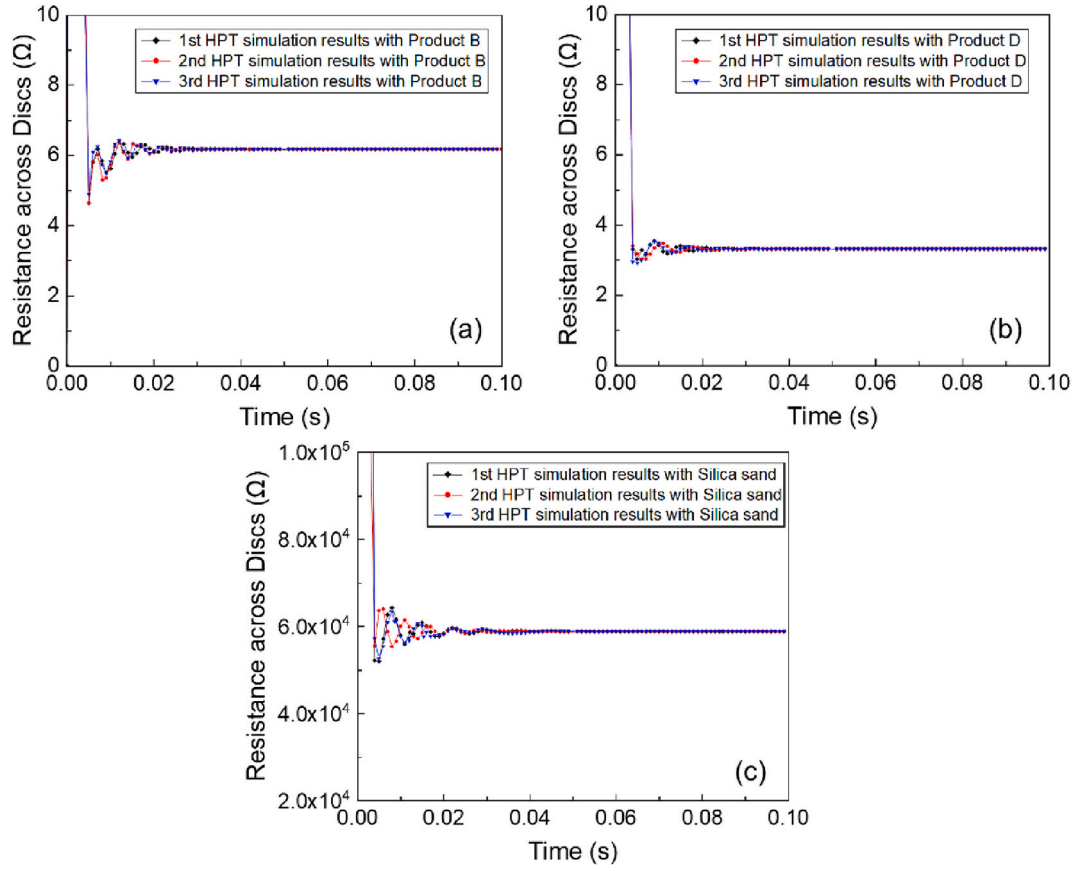


Fig. 16. Numerical results of electrical resistance for the HPT system with over-application of different materials: (a) Product B, (b) Product D, and (c) Silica sand.

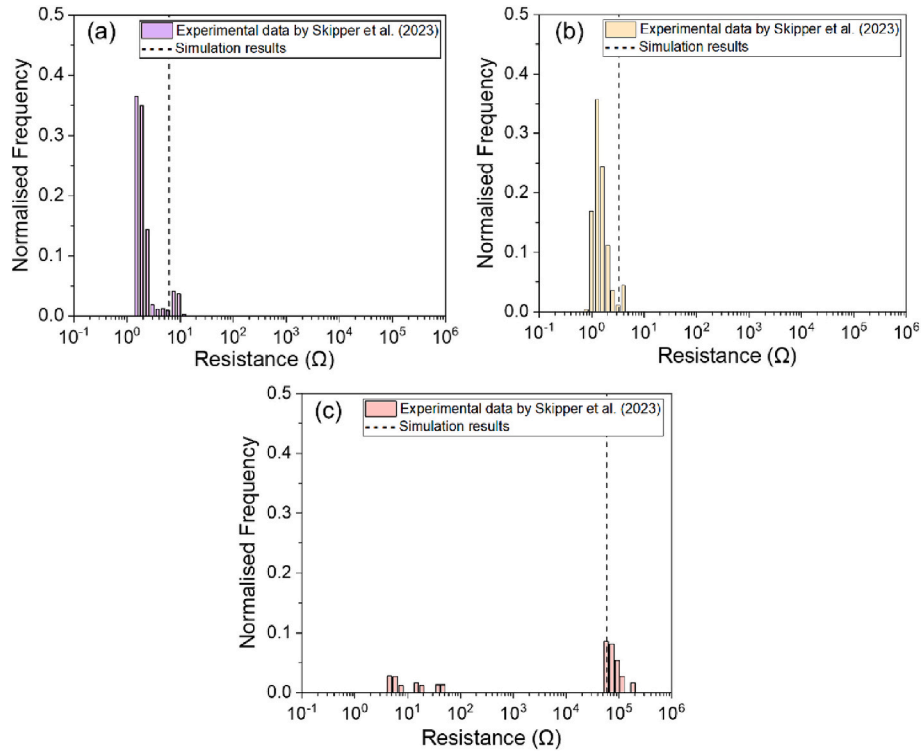


Fig. 17. Comparison between numerical predictions and the HPT experimental outputs: (a) Product B, (b) Product D, and (c) Silica sand.

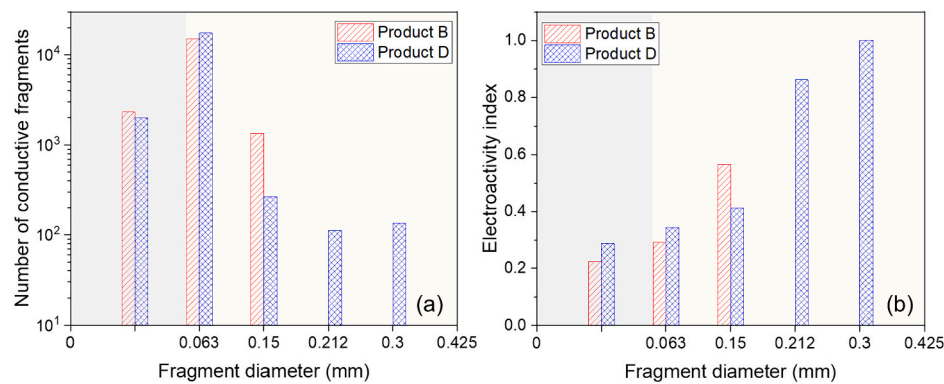


Fig. 18. (a) The number of electroactive fragments and (b) the degree of electrical activity of the fragments at each corresponding size for two conductive materials in the HPT test (The X-axis values represent the test sieve aperture size following BS1377–2:1990 [58]). The light orange area indicates that fragments retained by the specified sieve size, whereas the light grey area shows fragments that pass through the smallest sieve size).

for particle-to-particle and particle-to-wall contacts, and validated using experimental data from tests involving a chain of stainless-steel particles and a steel particle bed. The proposed model was applied to an example case, where an HPT test was used to study the effect of three different materials on the electrical response of the system. The main contributions of this study can be summarised as follows:

- 1) For both particle-particle and particle-wall conductions, the electrical resistance decreases dramatically as the mechanical force starts to increase. As the mechanical force keeps increasing, the downward trend of the electrical resistance asymptotes to a constant value. Changes in the contact surfaces for both particle-to-particle and particle-to-wall contacts can directly influence the electrical resistance of the system.
- 2) In the metallic particle system, contact resistance contributes significantly more than bulk resistance of the particles. This is evident in the numerical calculation, which exhibits a similar trend to the elastic Hertz theory as the mechanical force increases.
- 3) Tracking the electrical conduction behaviour within a particulate system using the HPT test as an application example, the numerical calculations of electrical resistance fall within the range of experimental results. Two commercial conductive materials, Product B and D, can effectively enhance the electrical conduction between the wheel and rail specimens compared to silica sand. Moreover, the electroactivity index reveals that the large-sized fragments contribute more to the bulk electrical conduction.
- 4) The proposed model allows for further investigation of the electrical response of particulate systems with real particle size distributions. In addition, it is expected to perform reasonable predictions for the electrical conduction behaviour of granular systems consisting of a mixture of conductive and non-conductive materials.

CRediT authorship contribution statement

Chao Zhang: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation. **Sadaf Mar-amazonou:** Writing – review & editing, Supervision. **David Milledge:** Writing – review & editing, Supervision. **Sadegh Nadimi:** Writing – review & editing, Supervision, Resources, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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